

Pipelined Data Path. (Note)

→ pipelined data path is obtained by subdividing the single cycle data path into several stages and placing pipeline registers b/w these stages.

— Five stages of the pipelined data path.

① Instruction fetch

- PC gives address to instruction memory
- instruction is fetched
- output stored in IF/ID Register
- PC incremented by 4 $PC' \rightarrow PC + 4$

② Instruction decode

- Decode the instruction
- Read the register files
- pipeline register - ID/EX

③ Execution

- ALU performs arithmetic and logical operation

④ memory access

- Data memory is accessed for load (lw) and store instructions
- other instructions simply pass through this stage

⑤ Write back

- the result is write back to the register file.

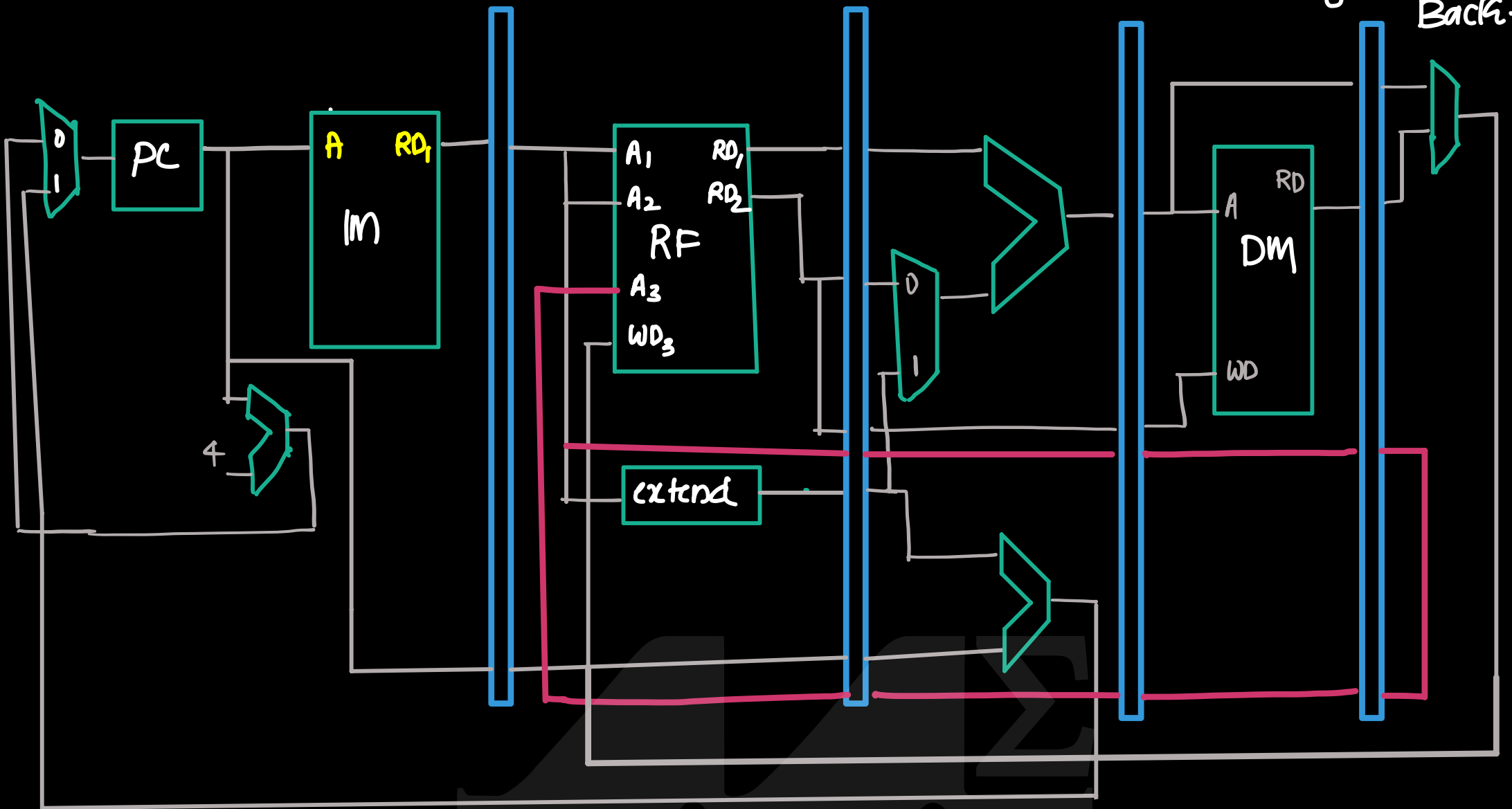
Fetch

Decode

execution

memory

Write Back.



pipe lined proucessor data path.

M SIGMA
GOKULAM

Pipelined Control

(N)

- pipelined processor uses the same control as the single cycle processor.
- so the control unit of pipelined processor is same as the single cycle processor.
- Refer Control signal for single cycle processor.

